

|              |                                                                                   |               |               |
|--------------|-----------------------------------------------------------------------------------|---------------|---------------|
| Date         | 30 September 2026                                                                 | Team ID       | SWTID-2026-26 |
| Project Name | Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer | Maximum Marks |               |

PROJECT DEVELOPMENT PHASE

Performance Test

Project team shall fill the following information in model performance testing template.

| S.No.                        | Parameter                    | Screenshot / Values                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                |                              |                                |                                |
|------------------------------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|------------------------------|--------------------------------|--------------------------------|
| 1.                           | Data Rendered                | <ul style="list-style-type: none"><li>• <b>Data Source:</b> ServiceNow ITSM operational database connected via Tableau Web Data Connector (WDC) extracting to a high-performance Tableau Data Engine Hyper Extract ( .hyper ).</li><li>• <b>Dataset Volume:</b> 12,450 procurement lifecycle records across requests, approvals, and tasks.</li><li>• <b>Relational Join Model:</b> Fact table <code>sc_req_item</code> joined with <code>sc_request</code> on Request ID, <code>sysapproval_approver</code> on Document ID and <code>sc_task</code> on Request Item.</li><li>• <b>Data Attributes Rendered:</b> REQ Number, RITM Number, Task ID (SCTASK), Catalog Item (Lenovo ThinkPad X1 Carbon), Approval Stat Assigned Group (Hardware), Task Short Description ("Laptop need to Configured"), Opened At, Approved At, Closed At, and Unit Price (\$1,100).</li></ul>                                                                                                                                                                                                                                                                                        |                                |                              |                                |                                |
| 2.                           | Data Preprocessing           | <ul style="list-style-type: none"><li>• <b>Tableau Prep Builder Transformations:</b><ol style="list-style-type: none"><li>1. <i>Imputation &amp; Null Handling:</i> Replaced null values in <code>Closed At</code> for active requests using <code>IFNULL([Closed At], NOW())</code>.</li><li>2. <i>Pruning Sandbox Data:</i> Filtered test orders ( <code>REQ0000001</code> to <code>REQ0000010</code> ) and non-hardware catalog requests.</li><li>3. <i>Datetime Splitting &amp; Standardization:</i> Formatted UTC timestamp fields into system datetime hierarchies for quarterly, monthly, and hourly analysis.</li><li>4. <i>Integrity Verification:</i> Validated 100% child task generation consistency for all manager-approved standard laptop requisitions.</li></ol></li></ul>                                                                                                                                                                                                                                                                                                                                                                        |                                |                              |                                |                                |
| 3.                           | Utilization of Filters       | <ul style="list-style-type: none"><li>• <b>Context Filter:</b> <code>[Catalog Item] = 'Standard Laptop'</code> (improves dashboard query performance and pins analysis to laptop workflow).</li><li>• <b>Dimension Filters:</b><ol style="list-style-type: none"><li>1. <i>Assignment Group Slicer:</i> Set default to <code>Hardware</code> queue with multi-group visibility.</li><li>2. <i>Approval State Slicer:</i> Multi-select for <code>Approved</code>, <code>Pending</code>, and <code>Rejected</code> states.</li><li>3. <i>Department Filter:</i> Dynamic selection across enterprise business units (Engineering, IT, Sales, Finance).</li></ol></li><li>• <b>Continuous Date Filter:</b> Interactive slider covering rolling 12 months with quick-filter presets (MTD, QTD, YTD).</li><li>• <b>Measure Filter:</b> Staging hours filter ( <code>[Staging Lead Time (Hours)] &lt;= 72</code> ) to isolate non-standard anomalous orders.</li></ul>                                                                                                                                                                                                    |                                |                              |                                |                                |
| 4.                           | Calculation fields Used      | <ul style="list-style-type: none"><li>• <b>Approval Latency (Hours):</b><br/><code>DATEDIFF('hour', [Approval Created At], [Approval Updated At])</code></li><li>• <b>Hardware Staging Duration (Hours):</b><br/><code>DATEDIFF('hour', [Task Created At], [Task Closed At])</code></li><li>• <b>Flow Automation Indicator:</b><br/><code>IF [Task Created By] = "system" OR CONTAINS([Task Short Description], "Laptop need to Configured") THEN "Automated via Flow" ELSE "Manual Ticket" END</code></li><li>• <b>SLA Compliance Status (≤ 24 Hours):</b><br/><code>IF [Hardware Staging Duration (Hours)] &lt;= 24 THEN "Within SLA" ELSE "SLA Breached" END</code></li><li>• <b>Flow Designer Automation Rate (%):</b><br/><code>SUM(IF [Flow Automation Indicator] = "Automated via Flow" THEN 1 ELSE 0 END) / COUNT([Task Number]) * 100</code></li></ul>                                                                                                                                                                                                                                                                                                    |                                |                              |                                |                                |
| 5.                           | Dashboard design             | <p><b>No of Visualizations / Graphs:</b> <b>6 Visualizations</b></p> <ul style="list-style-type: none"><li>• <b>1. KPI Summary Banner:</b> Big Numbers (BANs) for Total Orders (1,248), Avg Staging Time (4.2h), Automation Rate (100%), and SLA Compliance (98.4%).</li><li>• <b>2. Dual-Axis Combo Chart:</b> Monthly order submission volume (Bars) vs. Staging duration turnaround (Line).</li><li>• <b>3. Horizontal Bar Chart:</b> Configuration and deployment throughput per Hardware team technician.</li><li>• <b>4. Donut Chart:</b> Distribution of laptop requisitions by business department.</li><li>• <b>5. Bullet Graph:</b> Staging SLA fulfillment percentage measured against the 95% target threshold.</li><li>• <b>6. Stage Funnel Chart:</b> Stage progression (Order Submitted <math>\rightarrow</math> Approved <math>\rightarrow</math> Flow Triggered <math>\rightarrow</math> Tas Dispatched <math>\rightarrow</math> Closed Complete).</li></ul> <table><tr><td><b>1,248</b><br/>TOTAL ORDERS</td><td><b>4.2h</b><br/>AVG LEAD TIME</td><td><b>100%</b><br/>FLOW AUTOMATION</td><td><b>98.4%</b><br/>SLA COMPLIANCE</td></tr></table> | <b>1,248</b><br>TOTAL ORDERS   | <b>4.2h</b><br>AVG LEAD TIME | <b>100%</b><br>FLOW AUTOMATION | <b>98.4%</b><br>SLA COMPLIANCE |
| <b>1,248</b><br>TOTAL ORDERS | <b>4.2h</b><br>AVG LEAD TIME | <b>100%</b><br>FLOW AUTOMATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>98.4%</b><br>SLA COMPLIANCE |                              |                                |                                |
| 6.                           | Story Design                 | <p><b>No of Visualizations / Graphs:</b> <b>7 Visualizations across 5 Story Points</b></p> <ul style="list-style-type: none"><li>• <b>Story Point 1 (Baseline Bottlenecks):</b> Area chart illustrating operational delays and ticket backlog accumulation under manual handoff workflows.</li><li>• <b>Story Point 2 (Approval Latency):</b> Box-and-whisker plot highlighting manager authorization turnaround across organizational tiers.</li><li>• <b>Story Point 3 (Flow Automation Impact):</b> Side-by-side comparative bar chart demonstrating the immediate drop to zero coordination latency upon activating Flow Designer.</li><li>• <b>Story Point 4 (Hardware Queue Velocity):</b> Heatmap matrix mapping staging efficiency across technician shifts and weekdays.</li><li>• <b>Story Point 5 (Executive Procurement Synthesis):</b> Interactive summary dashboard correlating total asset expenditure with operational labor hours saved.</li></ul>                                                                                                                                                                                                |                                |                              |                                |                                |

